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# Author:Winfield, Date: 02/24/2025, Purpose: To test the ANOVA function

# Import ggplot2 for plotting
library(ggplot2)

# Importing dummy dataset
PATH <- "https://raw.githubusercontent.com/guru99-edu/R-Programming/master/poisons.csv"
df <- read.csv(PATH)

# View levels in the 'treat' column (to understand factor levels)
levels(factor(df$treat))

# OPTIONAL: Plot data
ggplot(df, aes(x = poison, y = time, fill = poison)) +
  geom_boxplot() +
  geom_jitter(shape = 15, color = "steelblue", position = position_jitter(0.21)) +
  theme_classic()

# Create hypothesis:
# Null: Treatment type does not significantly affect time
# Alternative: At least one treatment differs significantly

# Run ANOVA
anova_result <- aov(time ~ poison + treat, data = df)
summary(anova_result)
```